

Transition Analysis Toolkit

A Reproducible Analysis of Student Transition Pathways

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1 Introduction

2 Methodology

3 Results

3.1 Model Performance

Table 1: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

Model	CV ROC AUC	Test ROC AUC	Test Rows
Decision Tree — With Risk	0.7072	0.7664	295
Random Forest — Without Risk	0.7101	0.7603	295
Random Forest — With Risk	0.7095	0.7599	295
Decision Tree — Without Risk	0.6994	0.7557	295

3.2 Feature Importance

Rank	Research theme	Mean importance
1	Secondary mathematics preparation	0.2755
2	Composite risk measure	0.2381
3	Preparatory mathematics	0.2320
4	Previous university performance	0.1090
5	Secondary school achievement	0.1058
6	Student characteristics	0.0395

3.3 Feature Rankings

Rank	Feature	Models	Mean importance
1	risk_index	2	0.4762
2	has_preparatory_subject	4	0.1694
3	secondary_preparation_level_Other	4	0.1019
4	secondary_preparation_level_Standard	4	0.0950
5	has_previous_current_subject_attempt	4	0.0693
6	prior_preparatory_subject_pass	4	0.0532
7	previous_current_subject_failure	4	0.0397
8	international_student	4	0.0395
9	secondary_school_mark_band_70	4	0.0394
10	secondary_preparation_level_Ext1	4	0.0348
11	secondary_school_mark_band_90+	4	0.0288
12	secondary_preparation_level_Ext2	4	0.0257
13	secondary_school_mark_band_80-89	4	0.0228
14	secondary_preparation_level_Advanced	4	0.0182
15	secondary_school_mark_band_70-79	4	0.0148
16	prior_preparatory_subject_failure	4	0.0094

4 Discussion

5 Conclusion

A Supplementary Material